

1.

a.

- Let R_1 , R_2 , and R_3 , respectively, represent the rates of the given reactions, in the units $\frac{\text{mol}}{\text{L-min}}$. Let G , Py , L , and W , respectively, represent the concentrations of Glucose, Pyruvate, Lactate, and Waste Products, in the units $\frac{\text{mol}}{\text{L}}$.
- The rate expressions of the reactions are given as follows:

$$R_1 = k_1 G - (I)$$

$$R_2 = k_2 (Py) - (II)$$

$$R_3 = k_3 G^2 - (I)$$

- The system of ODEs governing the concentrations of the metabolites involved in the reactions is given as follows:

$$\frac{dG}{dt} = -k_1 G - k_3 G^2 \quad - \quad (1)$$

$$\frac{d(Py)}{dt} = k_1 G - k_2 (Py) \quad - \quad (2)$$

$$\frac{dL}{dt} = k_2 (Py) \quad - \quad (3)$$

$$\frac{dW}{dt} = k_3 G^2 \quad - \quad (4)$$

b.

- The system was simulated at 100 time points between $t = 0$ minutes and $t = 10$ minutes using the `solve_ivp()` function from the `scipy.integrate` sub-package in Python.

- The concentration profiles of all the species involved in the given bioreactor system, for the given parameters, are given as follows:

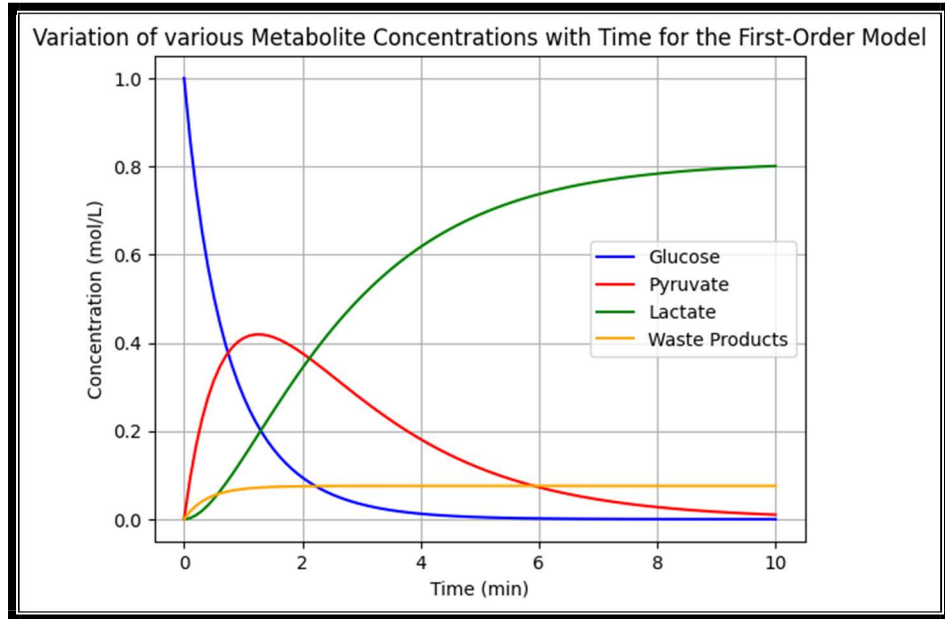


Figure 1: Concentration profiles for all the metabolites involved in the given bioreactor system, based on simulations using the First-Order Model

- Parameters:

- $k_1 = 1 \text{ min}^{-1}$
- $k_2 = 0.5 \text{ min}^{-1}$
- $k_3 = 0.2 \frac{\text{L}}{\text{mol-min}}$

- Initial Concentrations:

- $G = 1 \frac{\text{mol}}{\text{L}}$
- $Py = L = W = 0 \frac{\text{mol}}{\text{L}}$

c.

- As observed in Figure 1, Glucose Concentration steadily decays from its initial value to $0 \frac{\text{mol}}{\text{L}}$ asymptotically over time.
- Pyruvate Concentration initially increases before reaching a maximum, after which it asymptotically approaches a value of $0 \frac{\text{mol}}{\text{L}}$.
- Moreover, Lactate and Waste Products' Concentrations initially increase and, over time, saturate at non-zero values over time. After a relatively long period of time, the steady state concentration of Lactate is substantially

greater than that of Waste Products, indicating that most of the Glucose present initially in the bioreactor is eventually converted to Lactose, which is, therefore, a preferred end-product over the Waste Products.

- The steady state is achieved as Glucose and Pyruvate Concentrations decay to $0 \frac{mol}{L}$, as governed by the system of ODEs given above.

d.

- The modified rate expressions for all the three reactions, considering Michaelis-Menten Kinetics for R_1 with parameters K_m and v_{max} , are given as follows:

$$R_1 = \frac{v_{max} \cdot G}{k_M + G} - (I)$$

$$R_2 = k_2(Py) - (II)$$

$$R_3 = k_3G^2 - (III)$$

- The system of ODEs governing the concentrations of the metabolites involved in the reactions is given as follows:

$$\frac{dG}{dt} = - \frac{v_{max} \cdot G}{k_M + G} - (1)$$

$$\frac{d(Py)}{dt} = \frac{v_{max} \cdot G}{k_M + G} - k_2(Py) - (2)$$

$$\frac{dL}{dt} = k_2(Py) - (3)$$

$$\frac{dW}{dt} = k_3G^2 - (4)$$

- The system was simulated at 100 time points between $t = 0$ minutes and $t = 10$ minutes using the `solve_ivp()` function from the `scipy.integrate` sub-package in Python.

- The concentration profiles of all the species involved in the given bioreactor system, for the given parameters, are given as follows:

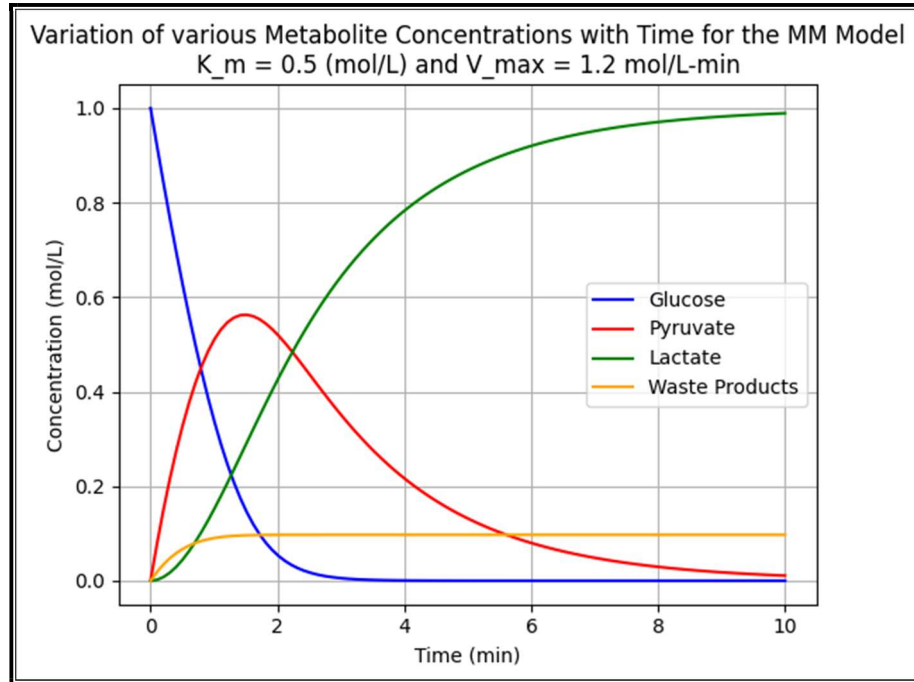


Figure 2: Concentration profiles for all the metabolites involved in the given bioreactor system, based on simulations using the Michaelis-Menten Model

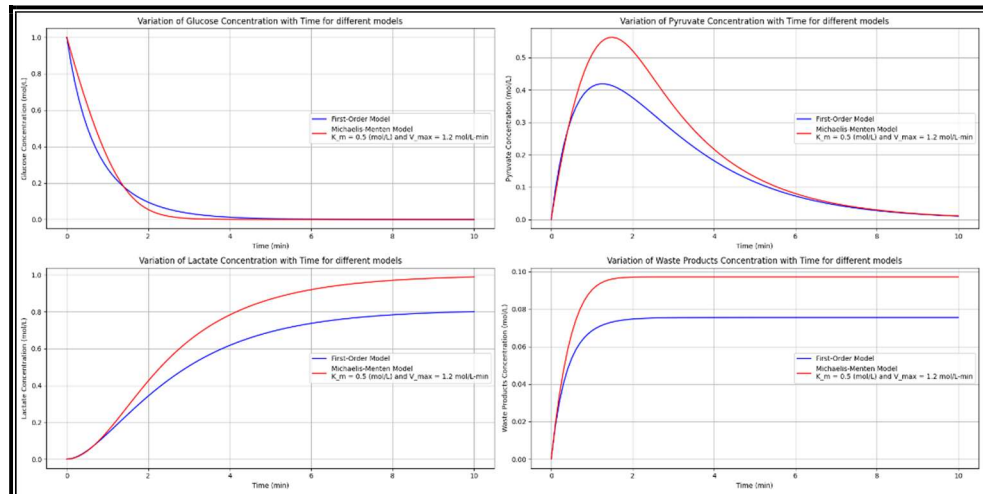


Figure 3: Comparison of the different concentration profiles of all the metabolites involved in the given bioreactor system, based on simulations using the First-Order and Michaelis-Menten models.

- Parameters:

1. $K_m = 0.5 \frac{\text{mol}}{\text{L}}$
2. $v_{max} = 1.2 \frac{\text{mol}}{\text{L} \cdot \text{min}}$
3. $k_2 = 0.5 \text{ min}^{-1}$
4. $k_3 = 0.2 \frac{\text{L}}{\text{mol} \cdot \text{min}}$

- Initial Concentrations:

1. $G = 1 \frac{\text{mol}}{\text{L}}$
2. $Py = L = W = 0 \frac{\text{mol}}{\text{L}}$

- We observe that Glucose Concentration steadily decays from its initial value to $0 \frac{\text{mol}}{\text{L}}$ asymptotically over time, in both cases. The rate of decrease is initially higher for the Michaelis-Menten Model, which is eventually overtaken by the First-Order Model.
- In both cases, Pyruvate Concentration initially increases before reaching a maximum, after which it asymptotically approaches a value of $0 \frac{\text{mol}}{\text{L}}$. However, the maximum Pyruvate Concentration achieved is higher in case of the Michaelis-Menten model as compared that achieved using the First-Order Model.
- In both cases, Lactate and Waste Products' Concentrations initially increase and, over time, saturate at non-zero values over time. After a relatively long period of time, the steady state concentration of Lactate is substantially greater than that of Waste Products, indicating that most of the Glucose present initially in the bioreactor is eventually converted to Lactose, which is, therefore, a preferred end-product over the Waste Products.
- However, the steady-state Lactate and Waste Products' Concentration achieved is higher in case of the Michaelis-Menten model as compared that achieved using the First-Order Model.

e.

- The concentration profiles of different metabolites for different values of the Michaelis-Menten parameter K_m are given as follows. The system was simulated at 100 time points between $t = 0$ minutes and $t = 10$ minutes using the `solve_ivp()` function from the `scipy.integrate` sub-package in Python.

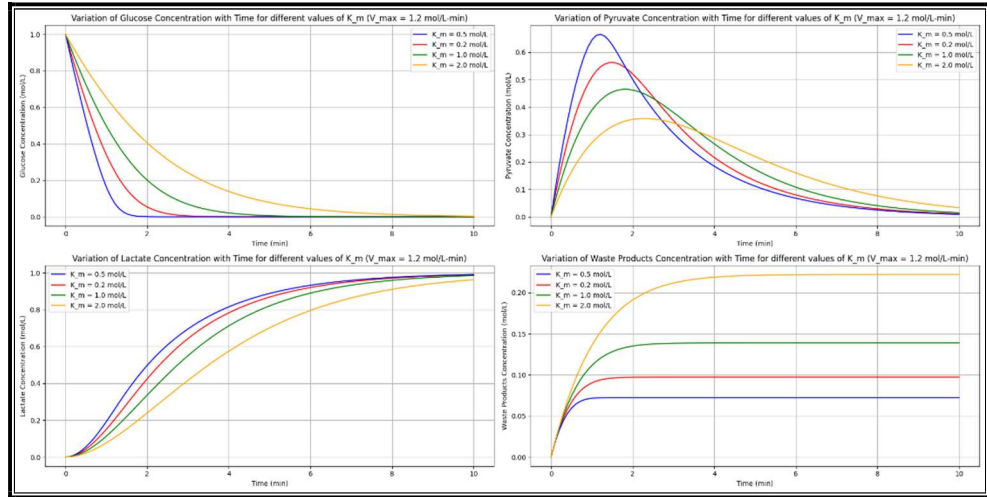


Figure 4: Comparison of the different concentration profiles of all the metabolites involved in the given bioreactor system, based on simulations performed for the Michaelis-Menten models.

- We observe that for higher values of K_m :
 1. The rate of decay of Glucose Concentration is relatively low.
 2. The rate of increase as well as that of decay of Pyruvate Concentration is relatively low. Moreover, the maximum Pyruvate Concentration is relatively low.
 3. The rate of increase of Lactate Concentration is relatively low.
 4. The rate of increase of Waste Products' Concentration is relatively high. Moreover, the steady state concentration of Waste Products is relatively high.
- Therefore, a higher value of K_m , or a lower affinity of the enzyme catalyzing reaction R_1 for its substrate Glucose, reduces the flux through the anaerobic respiration pathway, and increases the flux through the pathway that produces waste products (for example, aerobic respiration).

2.

a. The required network diagram is given as follows:

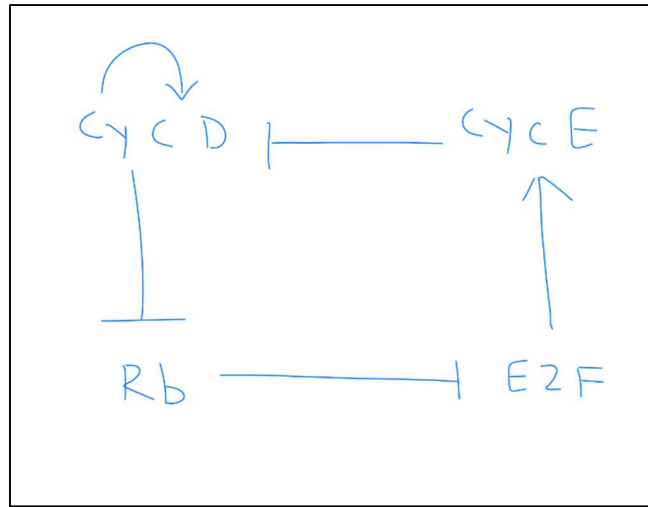


Figure 5: Network of Boolean Interactions

b. Let CycD, Rb, E2F, and CycE represent the Boolean Variables corresponding to the respective genes. The update rules, following a synchronous update strategy, are given as follows:

$$\begin{aligned}
 CycD(t+1) &= CycD(t) \text{ AND } (NOT(CycE(t))) \\
 Rb(t+1) &= NOT(CycD(t)) \\
 E2F(t+1) &= NOT(Rb(t)) \\
 CycE(t+1) &= E2F(t)
 \end{aligned}$$

c. The state transition table for all possible states is given as follows:

CycD(t)	Rb(t)	E2F(t)	CycE(t)	CycD(t+1)	Rb(t+1)	E2F(t+1)	CycE(t+1)
0	0	0	0	0	1	1	0
0	0	0	1	0	1	1	0
0	0	1	0	0	1	1	1
0	0	1	1	0	1	1	1
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	0
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	1
1	0	0	0	1	0	1	0
1	0	0	1	0	0	1	0
1	0	1	0	1	0	1	1
1	0	1	1	0	0	1	1
1	1	0	0	0	0	0	0
1	1	0	1	0	0	0	0
1	1	1	0	1	0	0	1
1	1	1	1	0	0	0	1

Figure 6: State Transition Diagram for the given Boolean Network

d. The attractors corresponding to all possible initial states are given as follows:

CycD(t = 0)	Rb(t = 0)	E2F(t = 0)	CycE(t = 0)	CycD(t = inf)	Rb(t = inf)	E2F(t = inf)	CycE(t = inf)
0	0	0	0	0	1	0	0
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	0
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	0	0	0	1	0	0
1	0	0	1	0	1	0	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	0	0
1	1	0	0	0	1	0	0
1	1	0	1	0	1	0	0
1	1	1	0	0	1	0	0
1	1	1	1	0	1	0	0

Figure 7: Attractor states for all possible initial states of the given Boolean Network

e.

- The system was simulated for all possible initial states by considering 20 update steps (an arbitrarily large number of updates). The results of the simulations are given as follows:

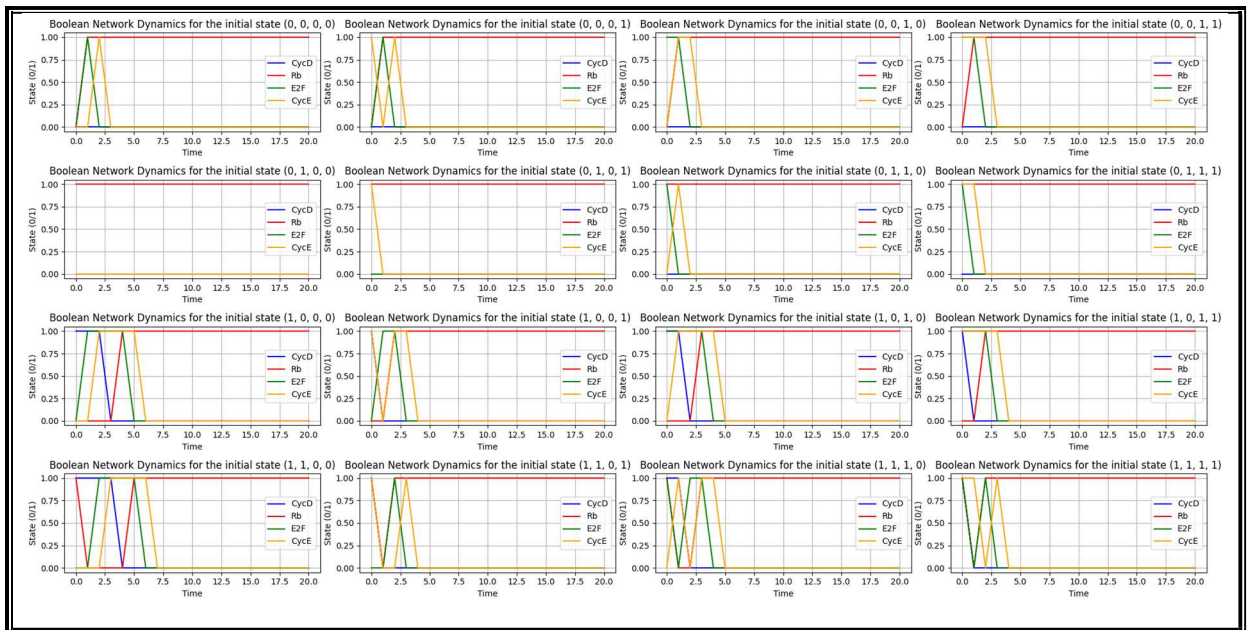


Figure 8: Boolean Network Dynamics for all possible initial states of the system, simulated for 20 update steps

- The given system has only one attractor state. Regardless of the initial state, the set of Boolean Variables eventually converges at the state (CycD, Rb,

$E2F, CycE = (0, 1, 0, 0)$. The only gene active in this state is Rb; the rest are in their “OFF” state.

- We observe that the steady state is achieved within less than 10 update steps; in some cases, the steady state is achieved in less than 5 update steps, indicating faster convergence, or higher proximity to the steady state along a set of hypothetical paths that connect each state to the one obtained by the system upon an update (the nodes and corresponding edges of this network are given by the State Transition Table given above in Figure 6).
- The states $(1, 0, 0, 0)$ and $(1, 1, 0, 0)$ require the maximum number of update steps (6 steps) to achieve the steady state.
- The state $(0, 1, 0, 1)$ requires the minimum number of update steps (1 step) to achieve the steady state.
- Thus, while all possible initial states eventually converge to a single attractor, the nature of the initial state determines the number of update steps required to achieve the steady state.